

Scientific Integrity and Research Misconduct

Maxim Borisyak, Claude Code, Andrey Ustyuzhanin

Constructor University Bremen

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Science as a Collective Enterprise

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Scientific is Iterative

Each result is an input to subsequent works:

- cited as established fact in future introductions;
- reused as a baseline for comparison;
- built upon in methods, proofs, and datasets;
- research is planned based on prior results.

Integrity violation $\implies$ systematic pollution;
 $\implies$ misguided future research;
 $\implies$ wasted resources and time.

The Spectrum of Misconduct

Category		Description
Honest error		Mistakes in analysis, code, misinterpretation
Questionable Research Practices (QRPs)		Decisions that inflate apparent evidence without outright fabrication
Research misconduct (FFP)		Deliberate fabrication, falsification or plagiarism

Fabrication and Falsification

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Fabrication

Fabrication: reporting data that were never collected.

Cases:

- **Hwang Woo-suk** (2004): fabricated human stem cell lines; two *Science* papers retracted;
- **Diederik Stapel** (2011): fabricated datasets for 55+ psychology papers;
- **Eric Poehlman** (2005): first US academic imprisoned for research fraud.

Falsification

Falsification: altering data, images, or analyses to misrepresent reality.

Cases:

- **Jan Hendrik Schön** (2002): reused figures across 16 papers in *Science* and *Nature*; identical noise patterns in independent experiments.

Plagiarism

Plagiarism: presenting others' intellectual work as one's own.

Forms:

- **verbatim copying:** reproducing text without proper attribution;
- **paraphrasing:** rewording ideas without citation;
- **idea theft:** using results from peer review or collaboration;
- **self-plagiarism:** republishing prior work without disclosure.

Copying via LLMs: using text produced by LLMs:

- LLMs can reproduce many texts almost verbatim (Ahmed et al. 2025, Huang et al. 2026):
- studies strongly suggests that LLM output is an amalgamation of others works;
- naturally, without attribution.

Consequences

- retraction — often years after publication;
- cascade retractions of papers that cited the work;
- loss of employment, funding, academic standing;
- criminal prosecution in extreme cases (Poehlman).

Retraction half-life in biomedicine: ≈ 10 years.

Statistical Malpractice

The Logic of Hypothesis Testing

A frequentist test controls the **false positive rate**:

$$P(\text{reject } H_0 \mid H_0 \text{ true}) = \alpha$$

This holds only if:

1. hypothesis fixed before seeing data;
2. test performed exactly once.

Both assumptions are routinely violated.

The Multiple Comparisons Problem

k independent tests under H_0 :

$$P(\text{at least one false positive}) = 1 - (1 - \alpha)^k$$

- $k = 1, \alpha = 0.05$: $P = 0.05$;
- $k = 20, \alpha = 0.05$: $P \approx 0.64$.

The Base Rate Fallacy

α controls $P(\text{reject } H_0 \mid H_0 \text{ true})$, not $P(H_0 \text{ true} \mid \text{reject } H_0)$.

Researchers routinely conflate the two — this is the **base rate fallacy**:

- $p < 0.05$ does **not** mean "5% chance the null is true";
- the probability that a significant result reflects a real effect depends on the prior probability that H_1 is true.

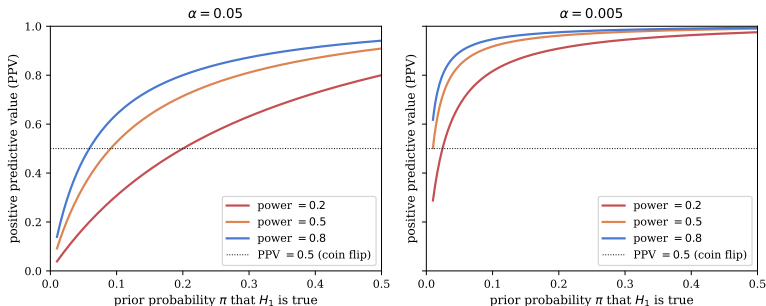
Positive Predictive Value

Let $\pi = P(H_1 \text{ true})$, $1 - \beta = \text{power}$. Probability that a significant result is genuine:

$$\text{PPV} = \frac{\pi(1 - \beta)}{\pi(1 - \beta) + (1 - \pi)\alpha}$$

- high power and high $\pi \Rightarrow$ most significant results are real;
- low π (exploratory research, many hypotheses) $\Rightarrow$ most are false positives, even at $\alpha = 0.05$.

Positive Predictive Value



PPV vs. prior π for three power levels. At $\alpha = 0.05$ (left), low-prior research produces mostly false positives even with 80% power. Lowering α to 0.005 (right) substantially improves PPV.

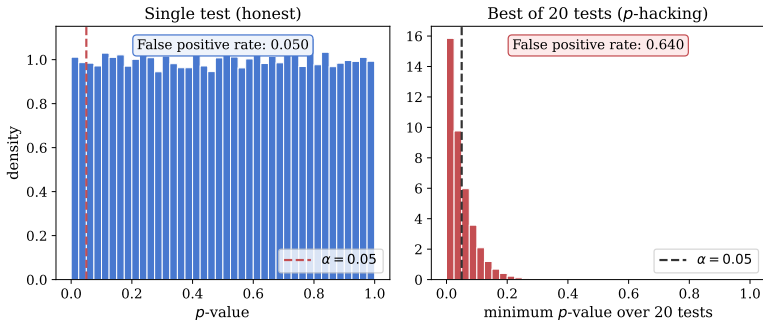
Why Most Published Research Findings Are False

Ioannidis (2005) extended the PPV framework to account for bias:

- in most fields, π is small — hypotheses are cheap to generate;
- studies are underpowered ($1 - \beta \approx 0.2\text{--}0.5$);
- bias (p -hacking, QRPs) inflates the effective false positive rate beyond α .

For many research areas, a significant result has less than even odds of reflecting a true effect.

p-Hacking in Practice



Single honest test: p -values uniform under the null (left). Best of 20 tests: false positive rate $\approx 64\%$ at nominal $\alpha = 0.05$ (right).

Forms of p-Hacking

Each decision below adds an implicit test:

- **optional stopping**: collecting data until $p < 0.05$, then stop;
- **covariate fishing**: changing features based on their effect;
- **test switching**: choosing different tests;
- **outlier sculpting**: removing some outliers but not others:
 - the definition of outliers is often tied to the model;
- **outcome selection**: testing many hypotheses, reporting significant ones.

The Garden of Forking Paths

Gelman & Loken (2014): at every step (preprocessing, outlier criteria, covariates, outcome definition) multiple legitimate choices exist.

Selecting the path that "works" after seeing data → searching over all paths.

The resulting p -value is not calibrated to α .

HARKing (Kerr, 1998): Hypothesising After Results are Known.

Mechanism:

1. collect data; explore; find a pattern;
2. rewrite the introduction as if the pattern was the original question.

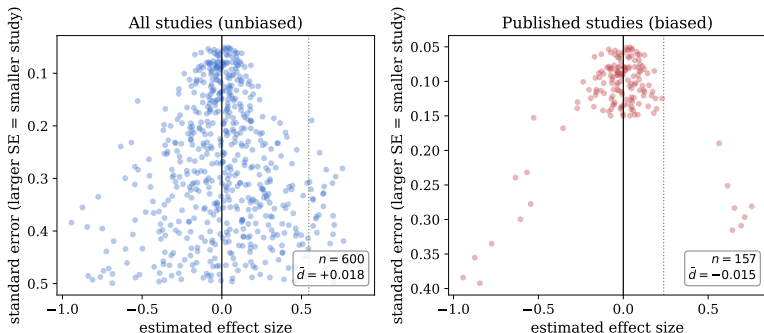
Effect: exploratory analysis presented as confirmatory $\Rightarrow$ evidential value destroyed.

Publication bias: positive results are published; null results are not.

Consequence: the literature overestimates effect sizes.

Funnel plot: studies scatter symmetrically around the true effect under no bias; publication bias creates visible asymmetry by removing small null-result studies.

Publication Bias



All studies (left) vs. published studies only (right). Small null-result studies are absent; mean reported effect $\bar{d}$ shifts away from zero.

Selective Outcome Reporting

Outcome switching: pre-register outcome A ; report outcome B when A is non-significant.

- Chan et al. (2004): 62% of clinical trials changed, introduced, or omitted a primary outcome between registration and publication;
- pre-registration creates an auditable record that makes switching detectable.

Manipulation of Methodology

Inappropriate Baselines

Reporting a method as better than an unfair comparison is not a contribution:

- **weak baseline:** poorly tuned or outdated comparison method;
- **task asymmetry:** easier formulation for the baseline;
- **metric cherry-picking:** reporting the metric where the proposed method wins.

Tuning one method on validation data while using defaults for the baseline invalidates the comparison.

Selective presentation of evidence:

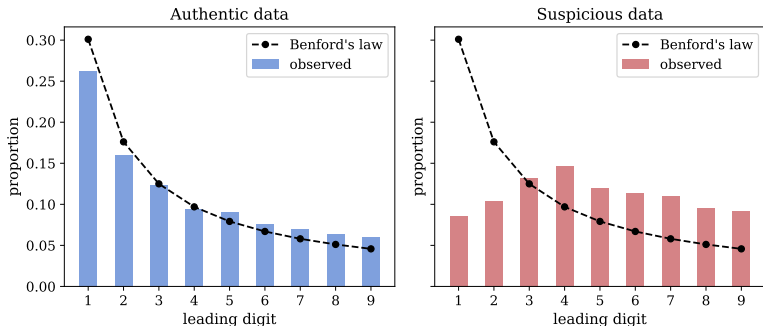
- reporting only conditions where the method succeeds;
- excluding runs with inconvenient results;
- evaluating on a curated subset of the test set.

Statistical effect: identical to p-hacking — reported performance does not generalise.

Fabrication Detection: Benford's Law

Benford's Law: naturally occurring data obey

$$P(\text{leading digit} = d) = \log_{10}(1 + 1/d).$$



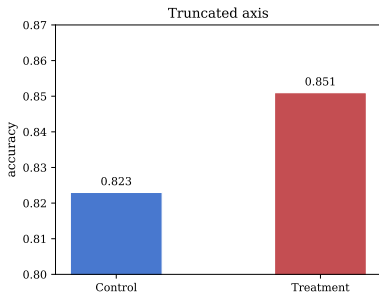
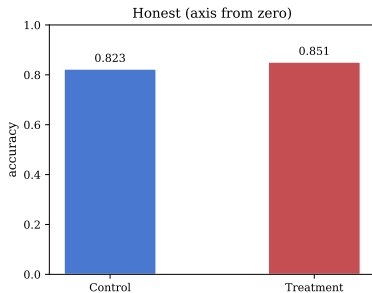
Authentic data (left) conforms to Benford's Law. Fabricated data (right) deviates — humans generating numbers tend toward uniformity.

Other Fabrication Signatures

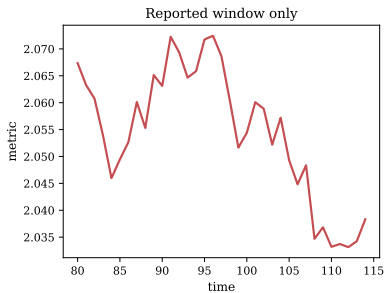
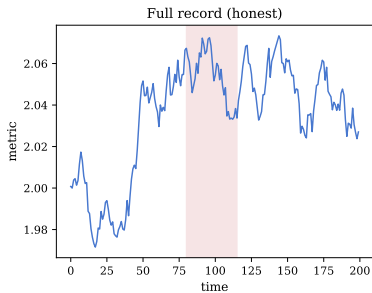
- **terminal digit preference:** fabricated data clusters on 0 and 5;
- **GRIM test:** certain means are arithmetically impossible for integer-valued scales.

Graphical Misrepresentation

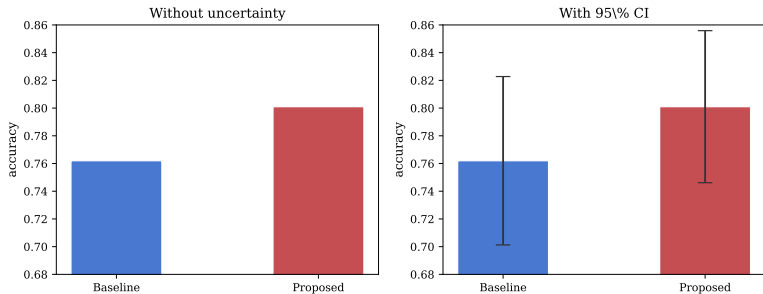
Truncated Y-Axis



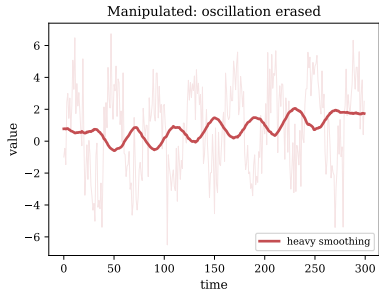
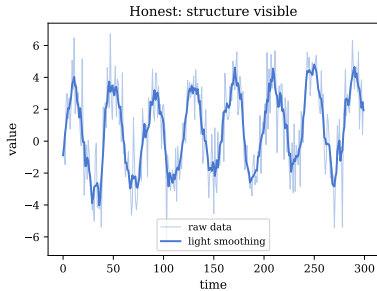
Cherry-Picking a Time Window



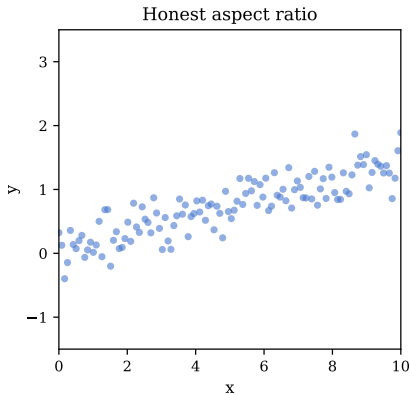
Omitting Uncertainty



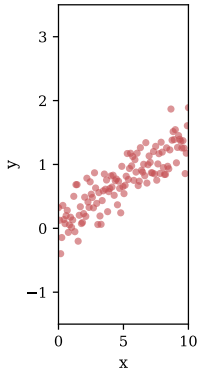
Over-Smoothing



Aspect Ratio Manipulation



Compressed x-axis: trend exaggerated



Machine Learning–Specific Issues

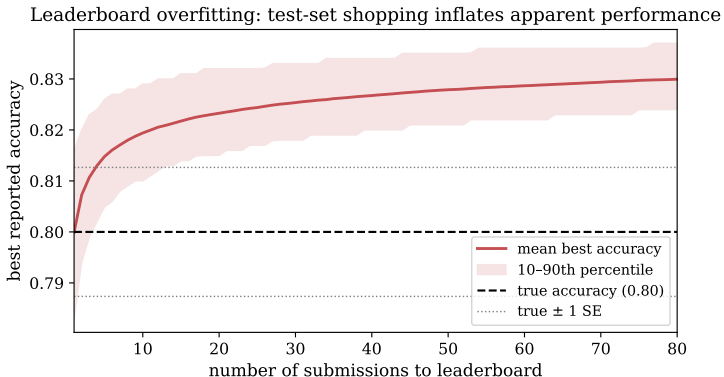
Test Set Contamination

Contamination: test information leaks into training or model selection.

Forms:

- test samples included in the training set (data leakage);
- preprocessing statistics (mean, variance) computed over the full dataset;
- future information leaks into the past in time-series settings.

Leaderboard Overfitting



Repeated evaluation against a fixed test set inflates best-so-far accuracy even when no model improves. True accuracy (dashed) is constant.

Leaderboard Overfitting: Analysis

k submissions, true accuracy p , test set size n , $\sigma = \sqrt{p(1-p)/n}$:

$$\mathbb{E} \left[\max_{i \leq k} \hat{p}_i \right] \approx p + \sigma \cdot \Phi^{-1} \left(\frac{k}{k+1} \right)$$

$n = 1000$, $p = 0.80$, $k = 80$:

- best-reported accuracy ≈ 0.815 .

Benchmark Gaming

- **dataset-specific tuning:** hyperparameters optimised for the benchmark, not by cross-validation;
- **metric shopping:** report only the metric where the method ranks first;
- **benchmark construction bias:** benchmark designed to favour properties of one's own method;
- **selective comparison:** omit concurrent work that would narrow the gap.

Pineau et al. (2021): a substantial fraction of ML papers do not release code, do not specify seeds, do not report variance, do not document hardware.

Henderson et al. (2018): deep RL results vary by up to 300% across seeds and implementation details — never disclosed in the original papers.

Publication Malpractice

Salami slicing: dividing one body of work into multiple minimal publications.

- inflates publication count without proportional contribution;
- fragments the record — reviewers cannot see the full picture.

Duplicate Publication

Duplicate publication: same work submitted to multiple venues without disclosure.

Legitimate:

- conference → journal with substantial extension (disclosed);
- preprint → peer-reviewed venue (standard practice).

Not legitimate: simultaneous submission, superficial repackaging without disclosure.

Authorship Fraud

ICMJE criteria: each author must have made a substantial intellectual contribution, participated in drafting, approved the final version, and be accountable.

Violations:

- **gift authorship:** non-contributors added for prestige;
- **ghost authorship:** real contributors omitted (common in industry-funded trials);
- **coercive authorship:** authorship demanded in exchange for resources.

The Replication Crisis

Scale of the Problem

Study	Rate	Field
Open Science Collaboration (2015)	36–39%	Psychology
Camerer et al. (2018)	62%	Economics
Begley & Ellis (2012)	11%	Cancer biology
Reprod. Project: Cancer Biology (2021)	50%	Cancer biology

Structural Causes

The crisis is driven by incentives, not primarily by fraud:

- **publication bias:** positive results published; null results filed away;
- **underpowered studies:** small n inflates estimates conditional on significance;
- **researcher degrees of freedom:** legitimate choices collectively inflate false positive rate;
- **career incentives:** novel positive results rewarded; replications are not.

Conditioning on $p < \alpha$ in an underpowered study ($1 - \beta \ll 1$):

$$\mathbb{E}[\hat{\delta} \mid p < \alpha] > \delta_{\text{true}}$$

Published estimates are biased upward. Adequately powered replications find smaller effects — the published literature is a biased sample of all studies conducted.

Safeguards

Registered Reports

Registered Reports: peer review before data collection; acceptance contingent on question and method, not outcome.

Standard	Registered Report
Idea → Data → Paper → Review	Idea → Review → Data → Paper
Acceptance depends on results	Acceptance independent of results
Publication bias acts	Publication bias structurally eliminated

- enables independent verification;
- allows error detection before findings propagate;
- makes the methods section verifiable, not merely aspirational.

Statistics:

- pre-specify primary outcome and analysis plan;
- apply multiple comparison corrections;
- report all outcomes.

Machine learning:

- fix splits before any modelling decision;
- multiple seeds (5-10); compute estimate errors;
- tune all baselines equally;
- release code.

When reading a paper:

- are baselines recent and competitively tuned?
- are null or negative results present?
- is the method section sufficient for reimplementation?
- **are estimate errors reported?**

Conclusion

Summary

Category	Examples
Misconduct (FFP)	Fabrication, falsification, plagiarism
Statistical QRPs	<i>p</i> -hacking, HARKing, optional stopping, outcome switching
Methodological QRPs	Cherry-picking, inappropriate baselines, benchmark gaming
Publication QRPs	Salami slicing, gift authorship, duplicate publication
ML-specific QRPs	Leaderboard overfitting, test contamination, metric shopping
Safeguards	Pre-registration, open data/code, registered reports

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